STELLA (YUXUAN) JIANG

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EDUCATION

University of California, Irvine Expected Mar. 2027
B.S. in Computer Science | **GPA: 3.88/4.0** | Dean's Honor List
Coursework: Data Structures and Algorithms, Machine Learning, Database Management, Software Engineering

WORK EXPERIENCE

Software Engineer Intern, Applied AI | Alibaba Group (global public tech company) Jul. 2026 - Sep. 2026
*CN | Tmall Campus digital services and retail ecosystem: **800+ universities and nearly 20M students.***

- Implemented venue booking with **PostgreSQL/SQLAlchemy transactions**, idempotency, and a FIFO waitlist, validating 20 concurrent booking requests with **zero overselling** and preserved queue ordering under retries.
- Implemented a **transactional outbox and leader leases** for booking workflows, integrating Prometheus metrics, structured logs, and request IDs to maintain observable event delivery during worker restarts and leader failover.
- Built user-profile services that transformed 100M raw events into **86.3M training records for ~1M users**, using **separate batch jobs** to isolate model training from request paths and preserve online profile-read performance at scale.
- Integrated React/TypeScript frontends with Python services through a **Node.js BFF**, using typed DTOs, swappable adapters, and **API contract tests** to enable parallel frontend development with mock services.
- Built a **Python RAG service** for campus knowledge search and an offline retrieval benchmark comparing BM25, BGE, and RRF on 6,562 queries; BGE achieved **76.4% Top-1 and 91.3% Top-3 accuracy**, guiding production retriever selection.
- Built reliability controls with a **deterministic Fact Guard**, request fingerprinting, and SQLite-backed atomic counters to avoid quota consumption by failed or invalid outputs; validated with 371 backend/smoke tests, browser tests, and 999-post regression.

Software Engineer, LLM Workflow | FessorAI, UCI Capstone Apr. 2026 - Jun. 2026
Irvine, CA | Full-stack AI education platform for structured textbook generation and in-page editing.

- Implemented a **six-stage textbook generation workflow** using **Django REST** and **PostgreSQL** to persist content states, execution progress, and intermediate outputs, while supporting four LLM providers through provider-agnostic interfaces.
- Designed **Celery/Redis-based orchestration** for long-running generation tasks with dependency-aware scheduling, retries, **durable progress tracking, and failure recovery**, validating reliability through 49 focused tests and fault-injection scenarios.
- Streamed text and tool events through **Django Channels, Redis, and WebSockets** into React content blocks for immediate navigation and review using a stable block contract; extended a **28-tool editing assistant** with native function calling and conversation memory.
- Resolved long-form editing failures by removing a **1,000-character truncation** and propagating **full-section context**.

Software Engineer Intern | NextTier Dec. 2025 - Feb. 2026
Sacramento, CA | Backend and evaluation services for a policy-grounded expense-audit copilot.

- Built an **AI expense-audit service** using **LangChain-based RAG**, OpenAI embeddings, document chunking, and vector search to generate policy-grounded approve/review/reject recommendations, reducing median review time from **6.4 to 2.1 min/report**.
- Designed a **FastAPI/Pydantic backend** separating stateless business services from stateful policy and receipt storage, with typed validation, error handling, structured logging, and health checks to improve reliability and observability.
- Implemented **LLM function calling** for internal expense API queries, cutting model calls **from 2.5 to 1.6 per audit** and token cost by **36%.**
- Containerized services with Docker for reproducible deployments and built **MLflow evaluation pipelines** comparing retrieval quality, decision accuracy, latency, and token cost before releases.

Founder & Software Engineer | FocusTrail Dec. 2025 - Present
Irvine, CA | UCI Startup Fund-supported local-first AI task execution coach; selected for Alibaba Tmall Campus OPC (5/50).

- Shipped a **local-first AI task execution architecture** across web and macOS/Windows prototypes, separating task planning, activity monitoring, model routing, and pre-inference privacy boundaries to keep raw activity data on-device before any model call.
- Implemented a Gemma-powered, **versioned task-DAG runtime** with dependency-aware replanning, completed-work protection, plan validation, state reconciliation, deterministic fallback, and undo to maintain execution consistency.
- Built React/Node.js/Express/SSE applications and **cross-platform activity agents** with task-aware classification, local/cloud model routing, privacy filtering, provider failover, and real-time feedback for model availability issues.

Software Engineer Intern | Join-Cheer (public enterprise software company, 1M+ users) Jun. 2025 - Sep. 2025
CN | Enterprise AI-agent platform and data services for government and financial-reporting workflows.

- Led a **Dify-to-Nuwa GPT migration** for an internal agent platform, preserving **legacy workflow compatibility** in production.
- Built **Python/Flask** services on ECS/OSS to normalize structured and unstructured data into **agent-callable tools**.
- Automated **Docker/Kubernetes CI/CD** releases across 10+ services, cutting **deployment time by ~40%** and eliminating manual rollbacks.
- Instrumented Function Compute with health checks and runtime metrics to surface post-migration failures across distributed services.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL
Frontend: React, Next.js, WebSocket, Server-Sent Events
Backend & Systems: Distributed Systems, REST APIs, FastAPI, Django, Flask, Node.js/Express, Celery
Databases & ORM: PostgreSQL, Redis, SQLite, SQLAlchemy
DevOps: Docker, Kubernetes, CI/CD, GitHub Actions, Alibaba Cloud, AWS, Prometheus
Testing: Pytest, Unit/Integration Testing, API Contract Testing, Regression Testing
AI & ML: RAG, Information Retrieval, Natural Language Processing (NLP), AI/LLM Agents, Function Calling, Structured Outputs, Model Evaluation